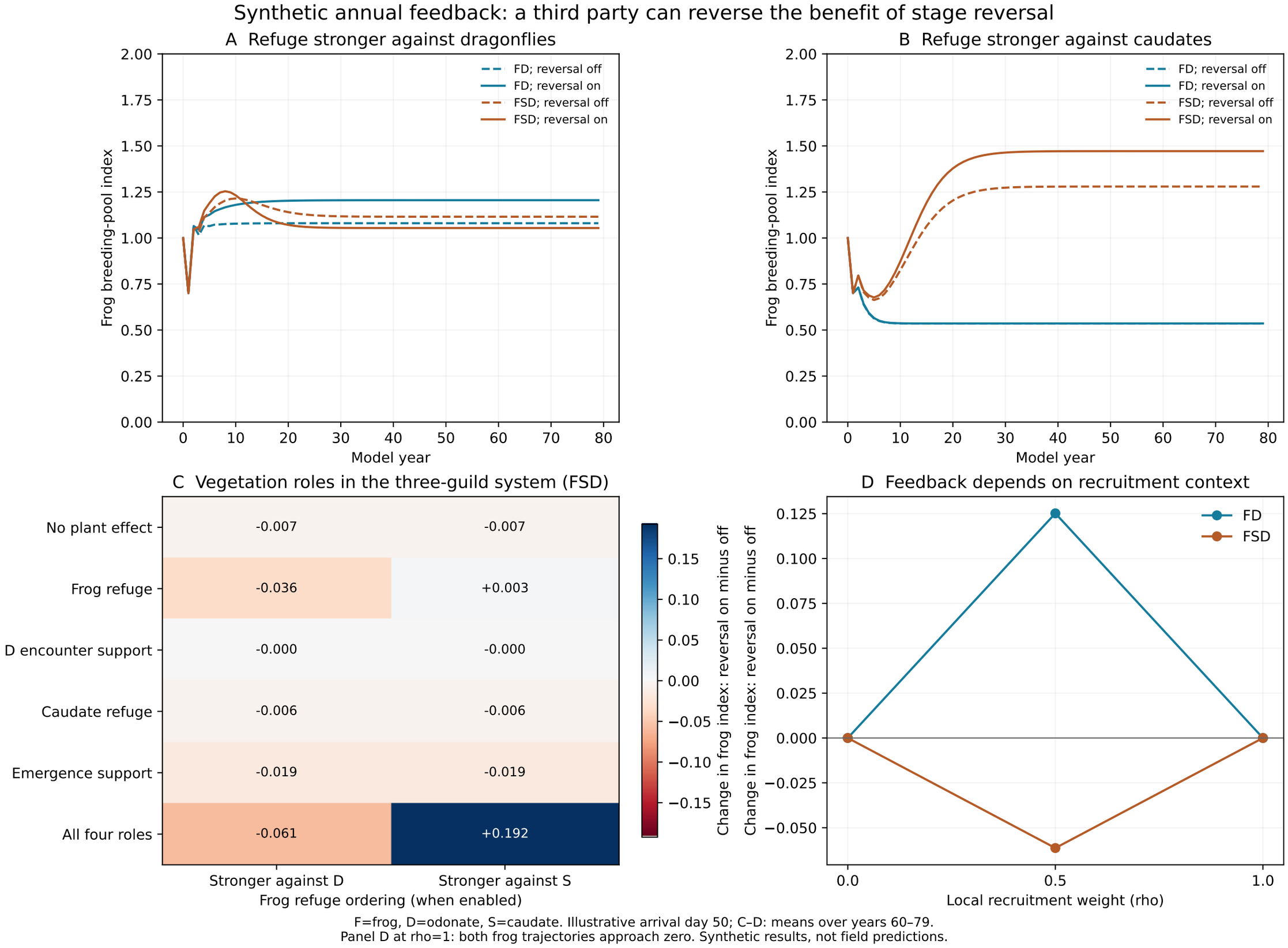
Synthetic annual feedback: a third party can reverse the benefit of stage reversal
A Refuge stronger against dragonflies
B Refuge stronger against caudates
FD; reversal off
FD; reversal on
FSD; reversal off
FSD; reversal on
Frog breeding-pool index
Model year
C Vegetation roles in the three-guild system (FSD)
No plant effect
Frog refuge
D encounter support
Caudate refuge
Emergence support
All four roles
-0.007
-0.007
-0.036
+0.003
-0.000
-0.000
-0.006
-0.006
-0.019
-0.019
-0.061
+0.192
Stronger against D
Stronger against S
Frog refuge ordering (when enabled)
Change in frog index: reversal on minus off
D Feedback depends on recruitment context
FD
FSD
Local recruitment weight (rho)
F=frog, D=odonate, S=caudate. Illustrative arrival day 50; C–D: means over years 60–79.
Panel D at rho=1: both frog trajectories approach zero. Synthetic results, not field predictions.